



IMPLEMENTING FALCON CONTROLLERS AND MISSIONS FOR QUADROTORS

Suchitha Channapatna and Amikosh Dube

INTRODUCTION: SUCHITHA CHANNAPATNA

- Autonomy Division, AI/ML Group
- Entering MS/PhD in Aerospace Engineering
 - MIT, Aerospace Controls Lab (ACL)
- BS in Aerospace Engineering
 - MIT, May 2026
 - Undergraduate research (including at ACL), WORMS (lunar robot)
- Hobbies: singing Carnatic Indian music, learning/teaching Sanskrit language, spending time with friends



INTRODUCTION: AMIKOSH (AVI) DUBE

- Autonomy Division – AI/ML Group
- Masters in ECE (Robotics)
 - Carnegie Mellon University, December 2026
- Bachelors in ECE
 - Purdue University, May 2025
- I have been doing quadcopter research for 6 years
 - Professor. Scherer (CMU)
 - Professor. Sundaram (Purdue)
- Hobbies: Basketball, Film Photos, Storm Chasing, and Astronomy

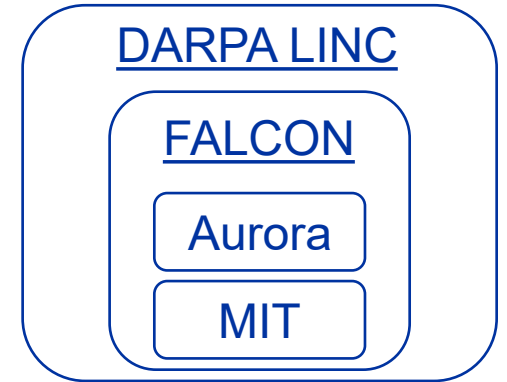


AGENDA

- 01 **FALCON Introduction**
- 02 **Suchitha's Work**
- 03 **Avi's Work**
- 04 **Lessons Learned**
- 05 **Q&A**

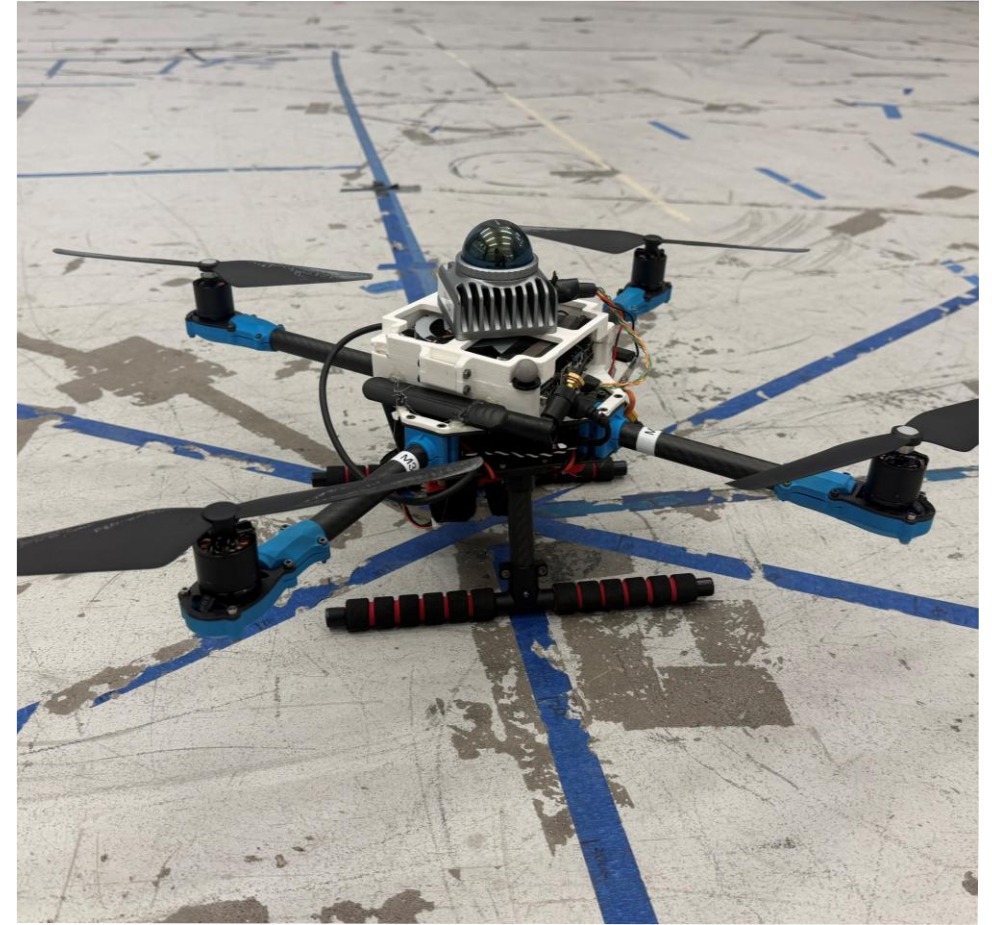
FAST ADAPTATION AND LEARNING FOR CONTROL ONLINE

- Adaptive control system to handle unpredicted events and prevent collisions at runtime
- Alleviate burden on operator during unexpected events
 - Currents in the river
 - Motor disturbances
 - FIFA Home Depot bucket



PROJECT GOALS

- Can FALCON be rapidly deployed on several different platforms?
- Intern focus: Quadcopters
 - Collaboration with MIT Aerospace Controls Lab
 - Prof. Jonathan How
 - Kresa Center for Autonomous Systems



SUCHITHA

Relative station keeping mission for quadcopters

SCOPE OF WORK

Recap across the internship

Workstream	Summary	Status
Extension to Multi-vehicle PX4 Sim	Extended ACL's trajectory generator module and PX4 stack to multi-vehicle simulation	✓ Complete
Visualization Tooling	Built Foxglove tools for Guidance Module and PX4 logs, Python animations/plots for CBF	✓ Complete
Station Keeping Guidance Module	Implemented velocity-based guidance law for relative station keeping	✓ Flying on HW
DARPA Demo	Live demonstration of AAOBLTR	✓ Complete
CBF Relative Station Keeping	Closed-form safety filter, validated in simulation and tested on hardware	✓ Flying on HW

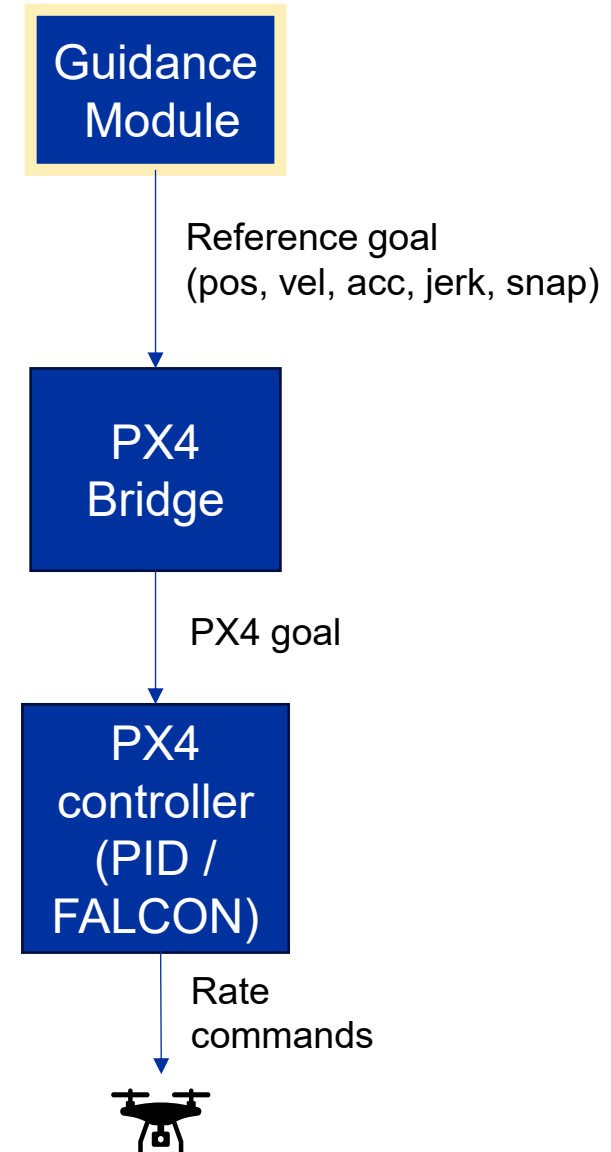
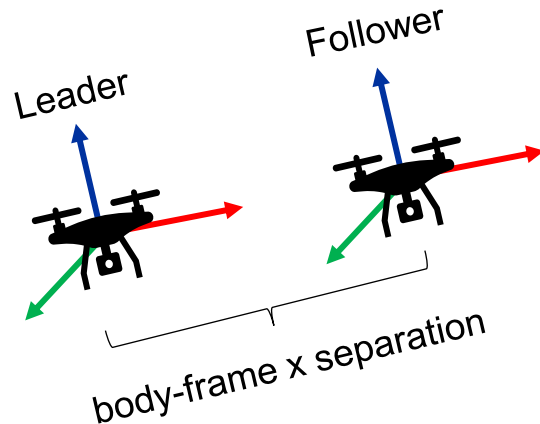
RELATIVE STATION KEEPING INTRODUCTION

Goal:

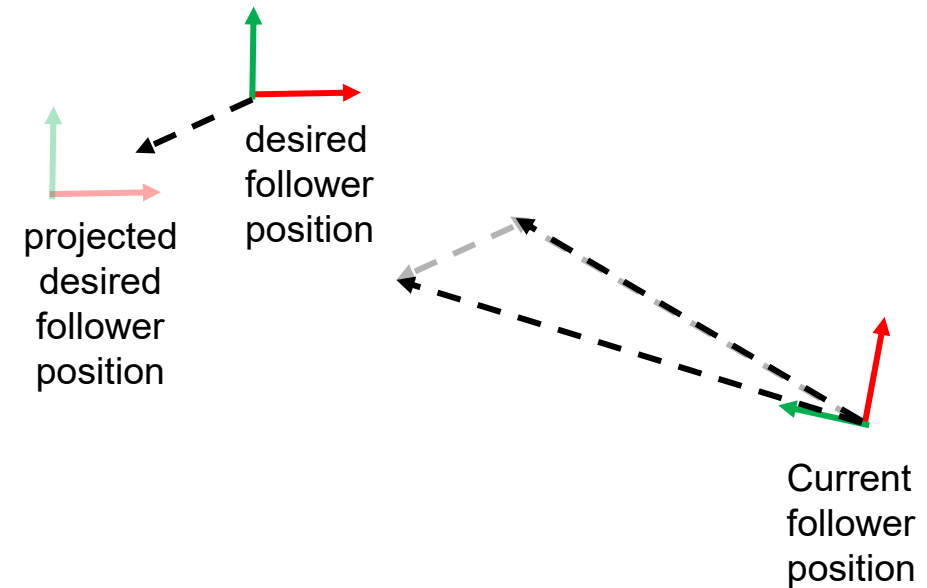
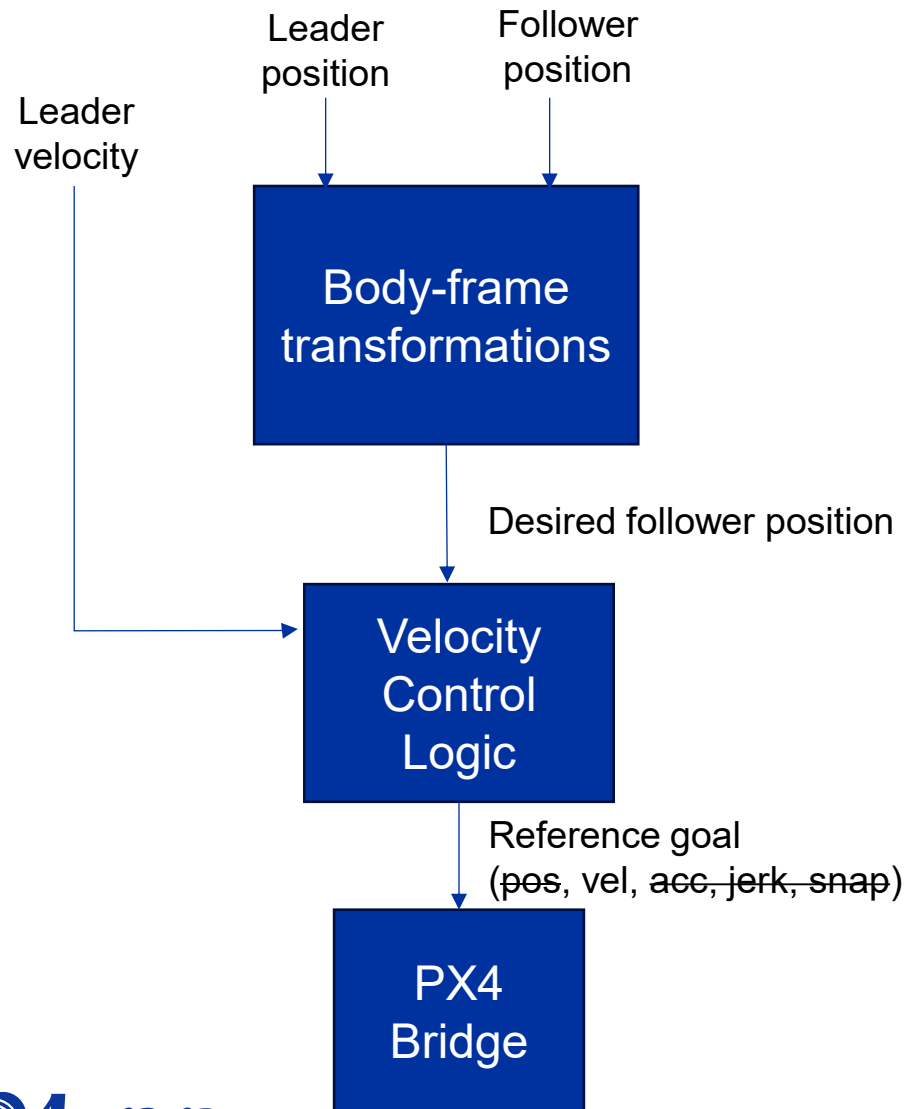
Maintain a **relative separation distance** in **any axes** with respect to the **leader vehicle's body frame**

Assumptions:

- Leader's body-x is aligned with yaw
- Follower's yaw always tracks leader's yaw
- Not responsible for deconflicting trajectories



RELATIVE STATION KEEPING APPROACH

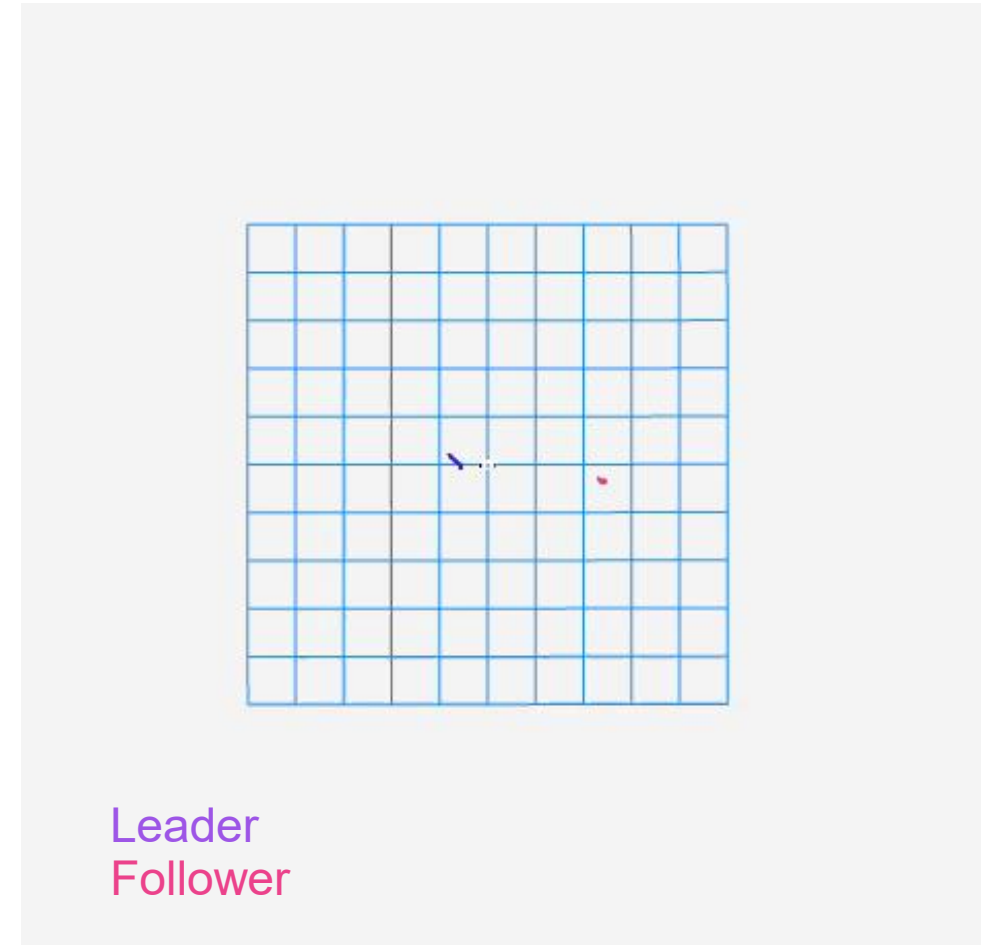
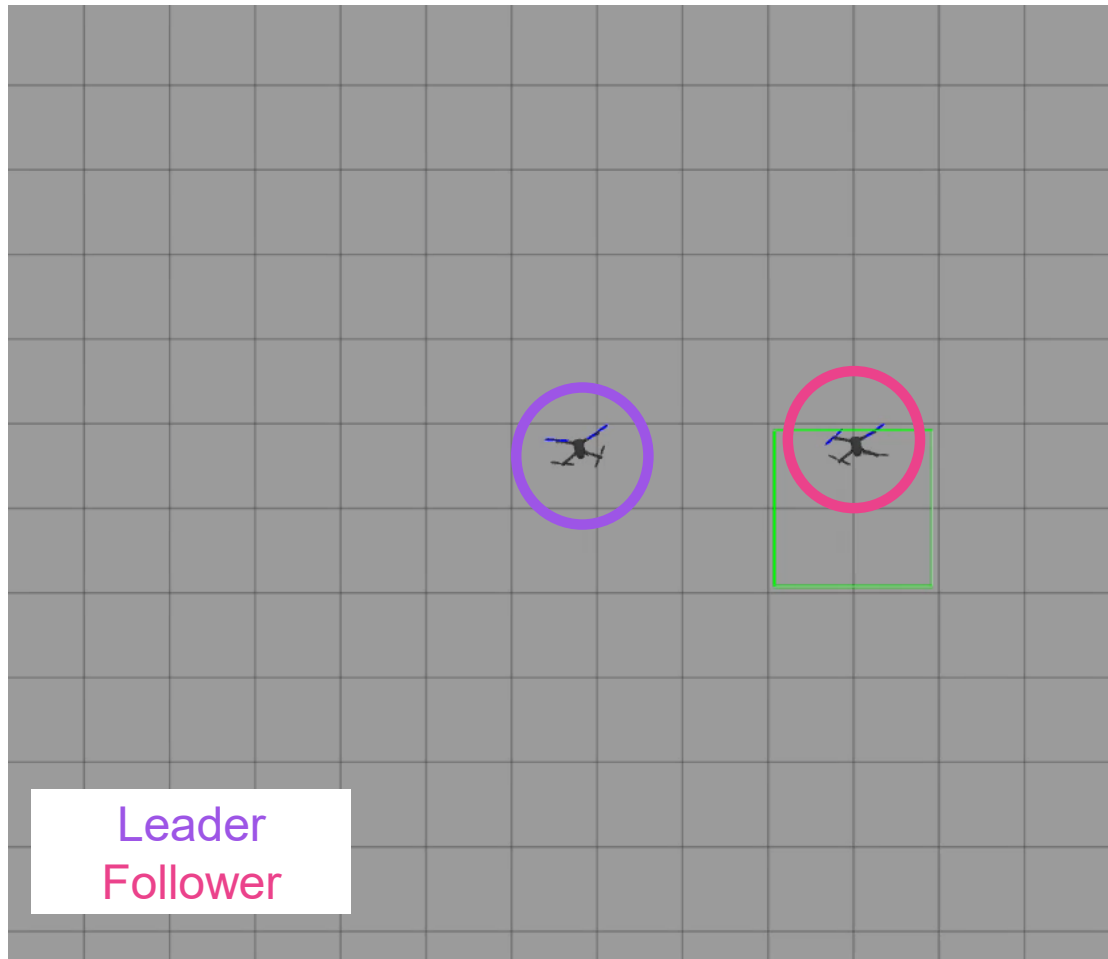


Velocity control law uses leader's velocity and a nonlinear term based on position error.

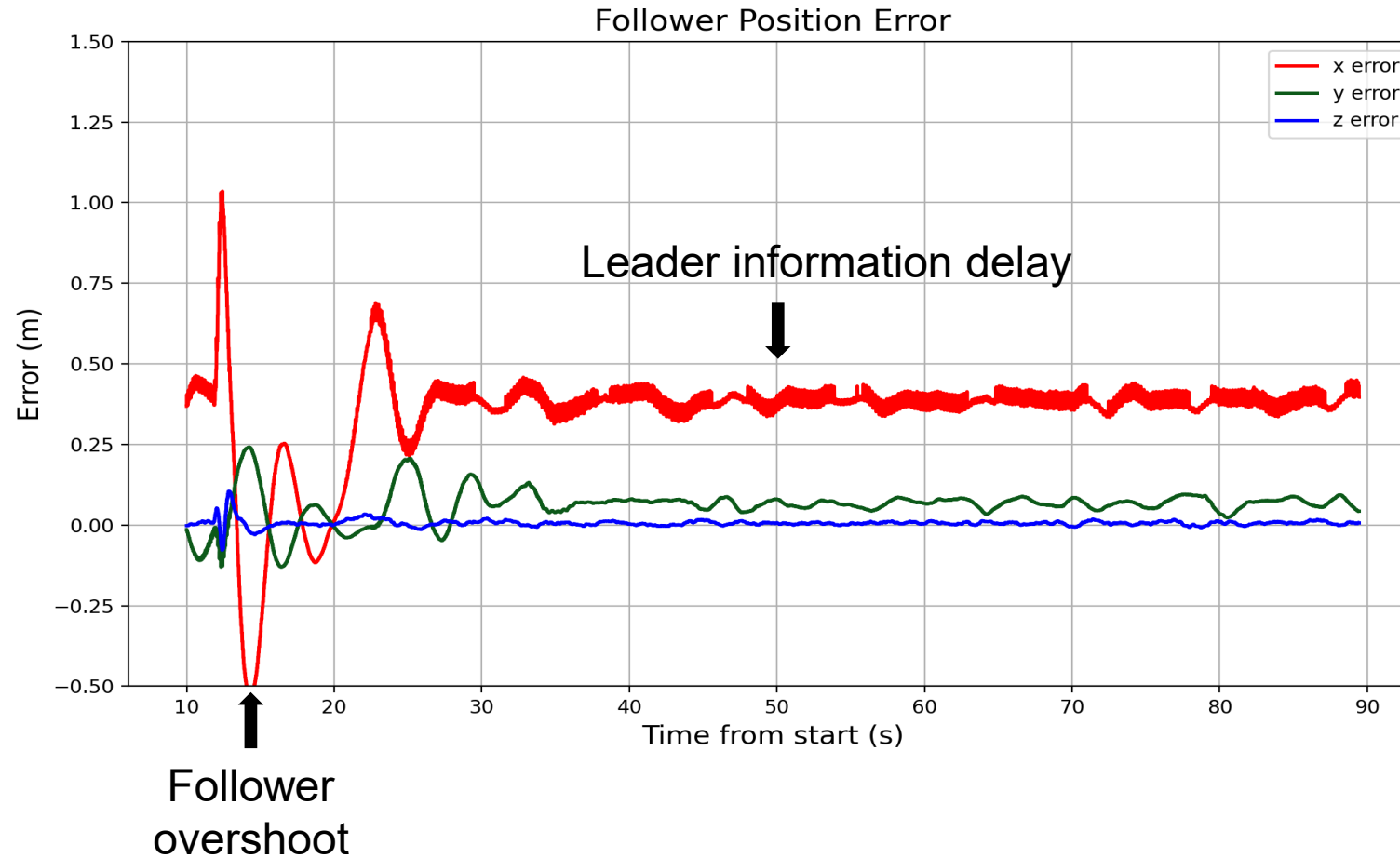
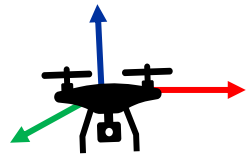
RELATIVE STATION KEEPING RESULTS - SIMULATION

Leader executes circular trajectory.

Follower maintains -3.0 meter separation in the leader's body-frame y-axis (**3 meters right**).



RELATIVE STATION KEEPING RESULTS - SIMULATION

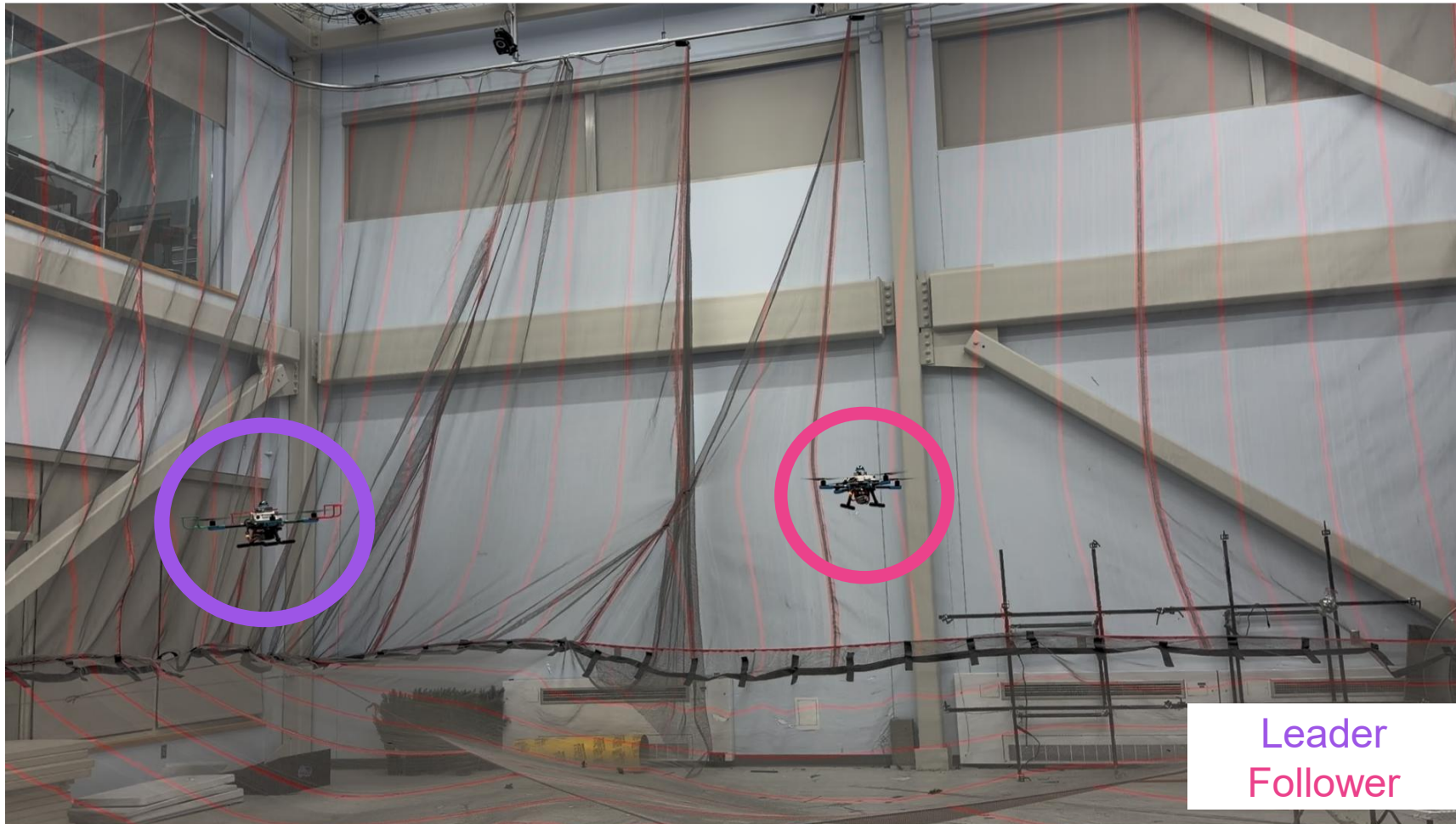


Velocity control law enables tracking. Highest error in x-axis (as expected based on leader trajectory).

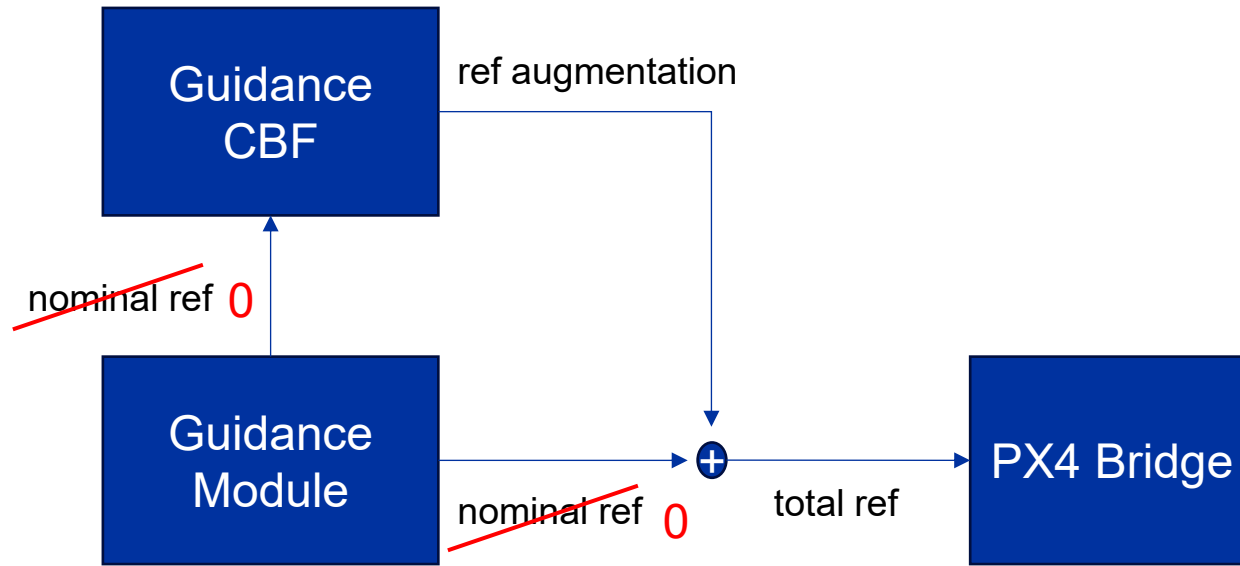
RELATIVE STATION KEEPING VIDEO - HARDWARE

Leader is hand-flown.

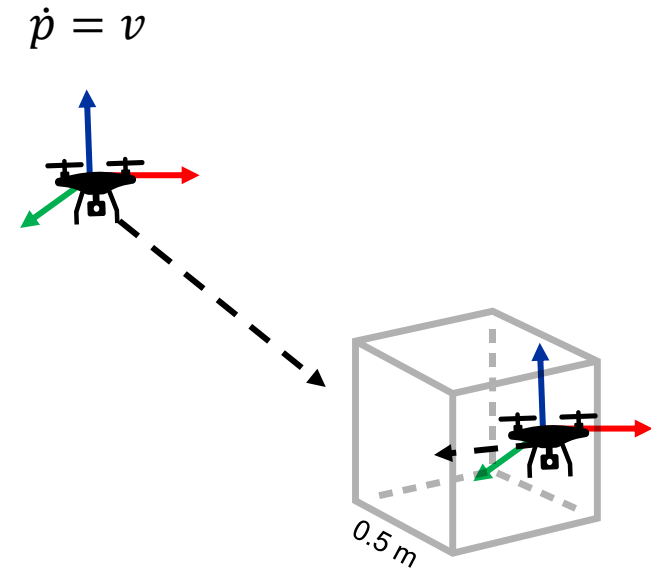
Follower maintains -3.0 meter separation in leader's body-frame y-axis (**3 meters right**).



CONTROL BARRIER FUNCTION - STATION KEEPING



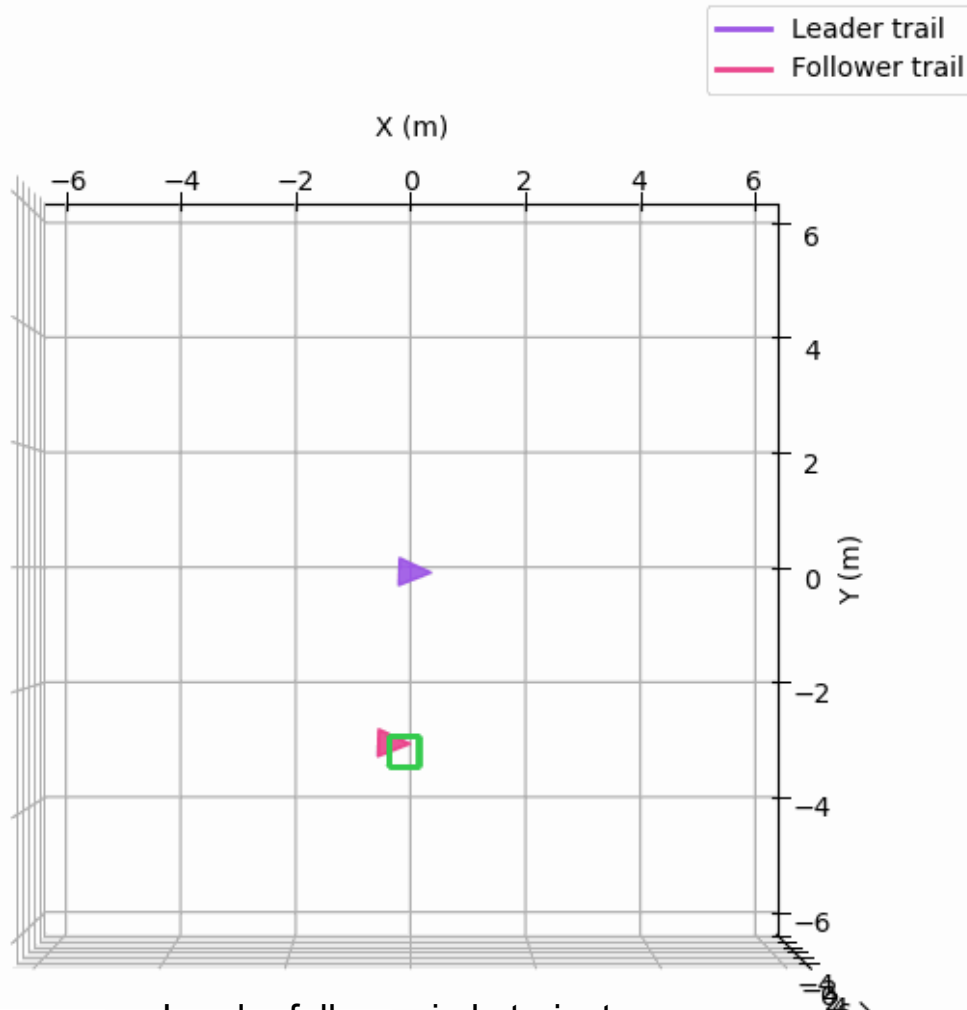
Analytical CBF solution possible for
box constraints



Safe set centered at follower's
desired position

CBF FOR STATION KEEPING RESULTS - SIMULATION

CBF Test Visualization | $t = 0.00$ s



CBF was successfully used for tracking, but follower did not remain within moving barrier during trajectory.

Potential reasons:

- Over-simplified model representation
- Tuning of CBF parameter
- Size of safe set
- Changing heading (better results with Line trajectory)

Leader follows circle trajectory.

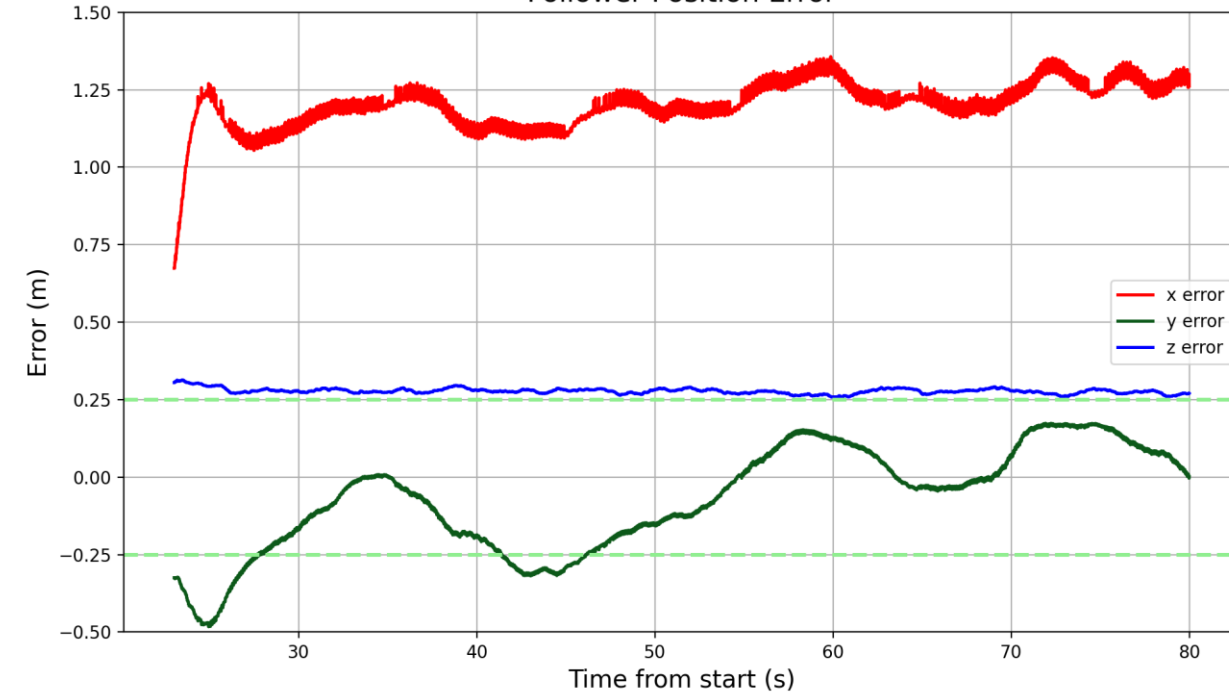
Follower tracks **3 m to the right**. No nominal velocity commands used (CBF only).



A Boeing Company

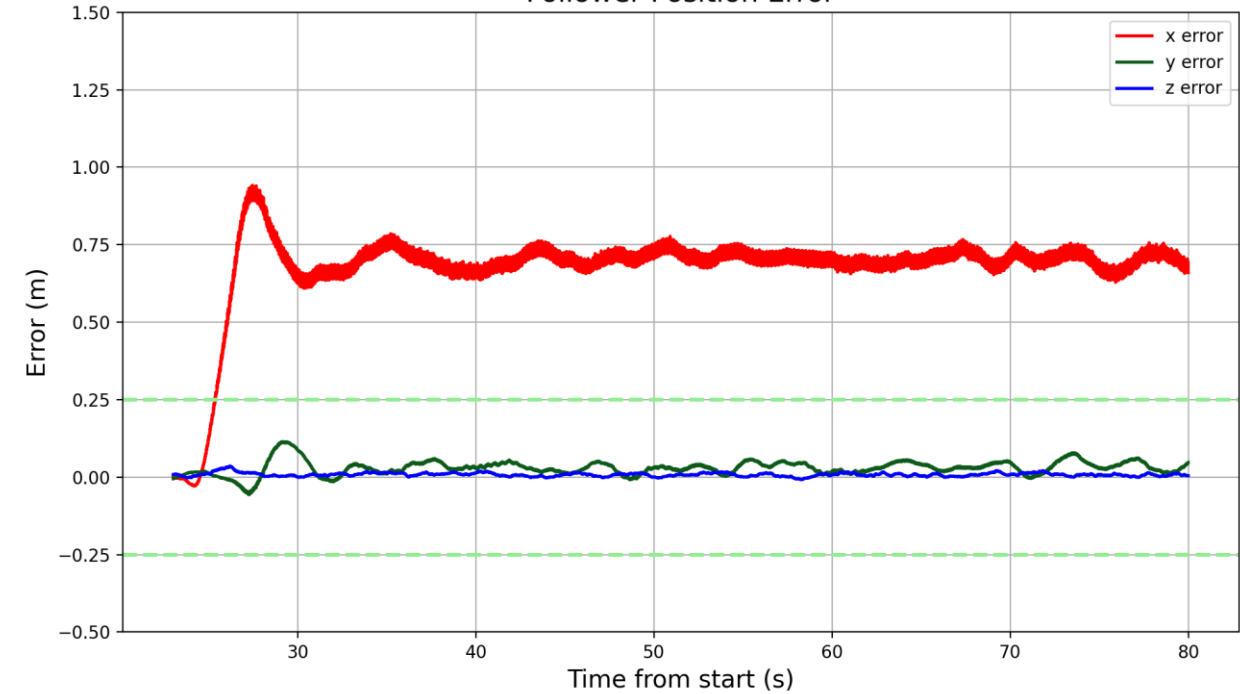
CBF FOR STATION KEEPING RESULTS - SIMULATION

Follower Position Error



CBF without nominal reference command

Follower Position Error



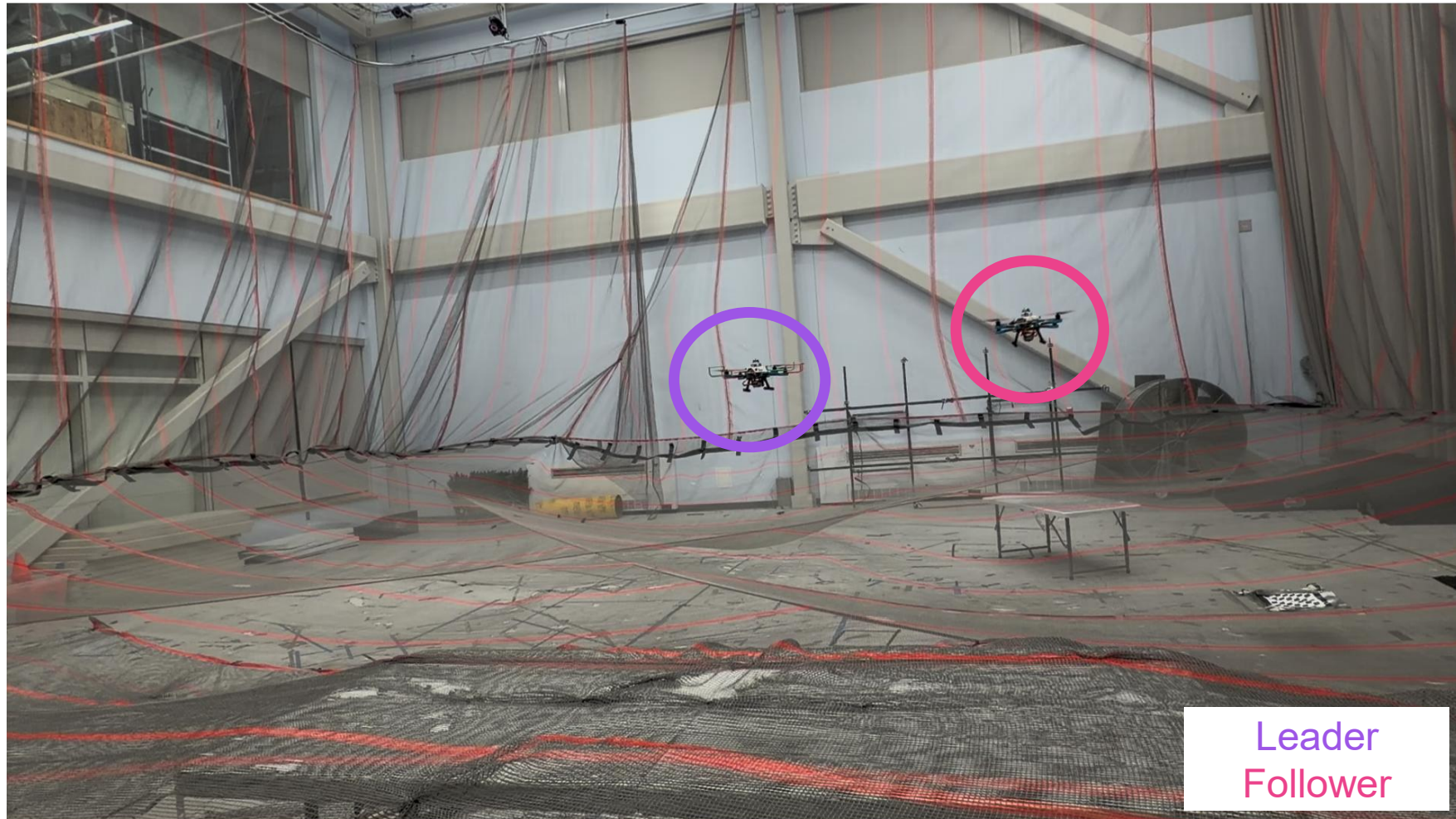
CBF with nominal reference command

x-error violates CBF constraint.
x-error reduces when nominal velocity command is used.

CBF FOR STATION KEEPING VIDEO - HARDWARE

Leader is hand-flown.

Follower maintains -3.0 meter separation in leader's body-y axis (**3 meters right**).



LESSONS LEARNED

Technical

- *Effectively* going down a PX4 rabbit-hole
- Sim → Real gap
 - Aligning simulation with hardware (and when it is unnecessary)
- Simplifying development and testing
 - Modular architectures and configuration files

Professional

- Preparing for a formal demonstration
 - Communication, automation, clarity of demo goals

AVI

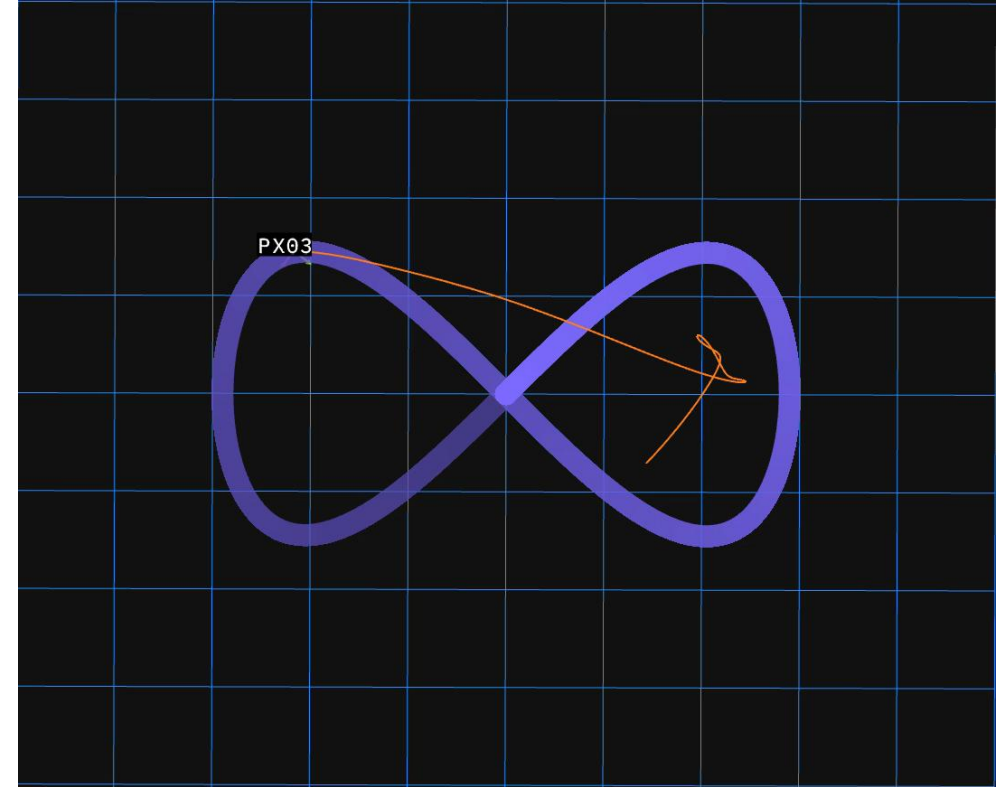
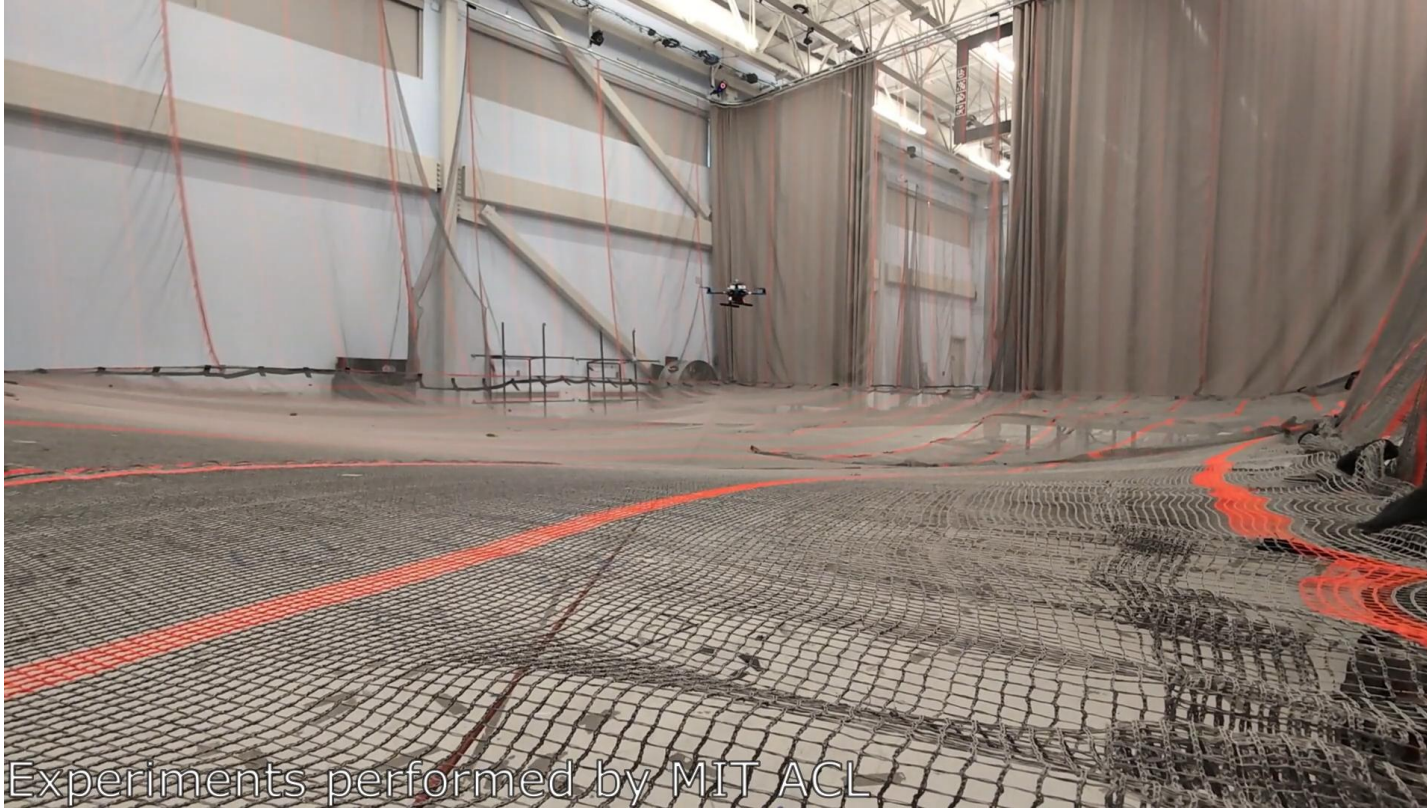
FALCON controllers implemented for quadcopters

SCOPE OF MY WORK

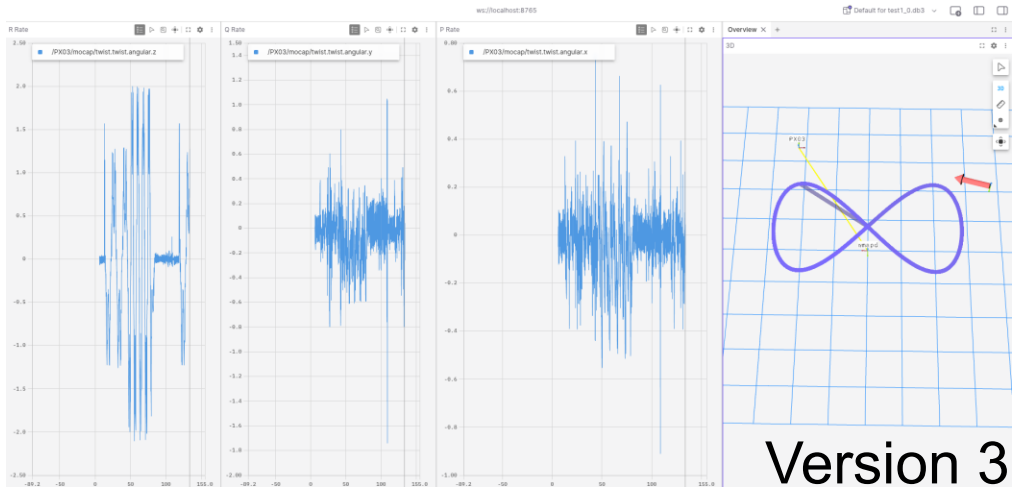
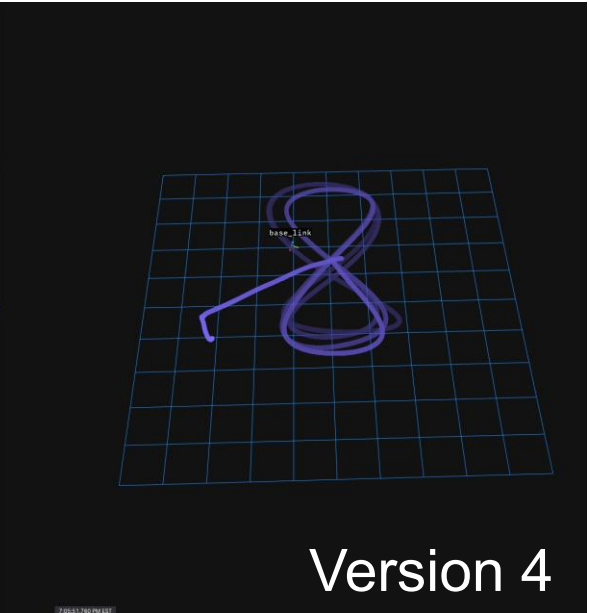
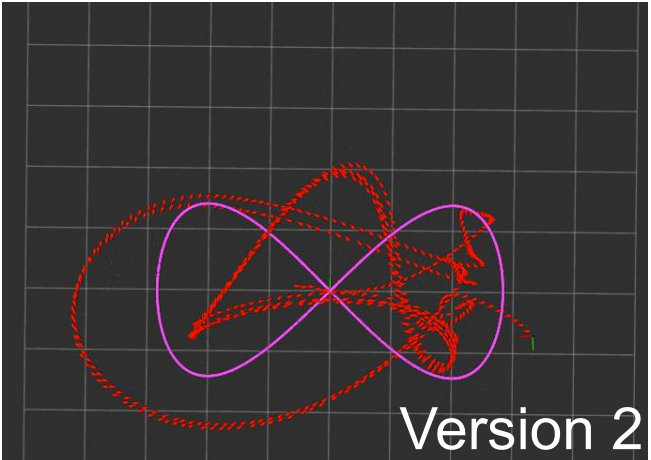
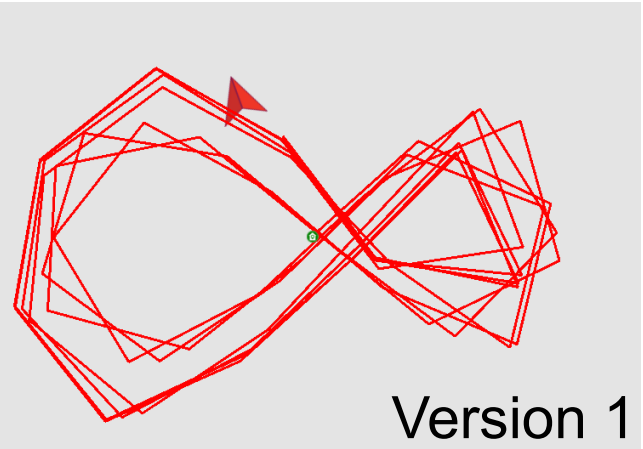
Recap across the internship

Workstream	Summary	Status
PX4 Debugging	Diagnosed and resolved issue in prior figure-8 implementation	✓ Fixed
Analysis Tooling	Built Foxglove tools, iterated RViz → full 3D replay	✓ Complete
Controller Library	Implemented PID, RSLQR, OBLTR, AAOBLTR behind a swappable interface	✓ Flying on HW
AAOBLTR Tuning	Tuned adaptive observer-based controller	✓ Stable flight
DARPA Demo	Live demonstration of AAOBLTR	✓ July 22
CBF Obstacle Avoidance	Closed-form safety filter, validated in simulation	⚠ Sim only — HW blocked
HLOC Perception	Improved boat detection / pose estimation pipeline	✓ Accuracy improved

DEBUGGING PREVIOUS FALCON IMPLEMENTATION

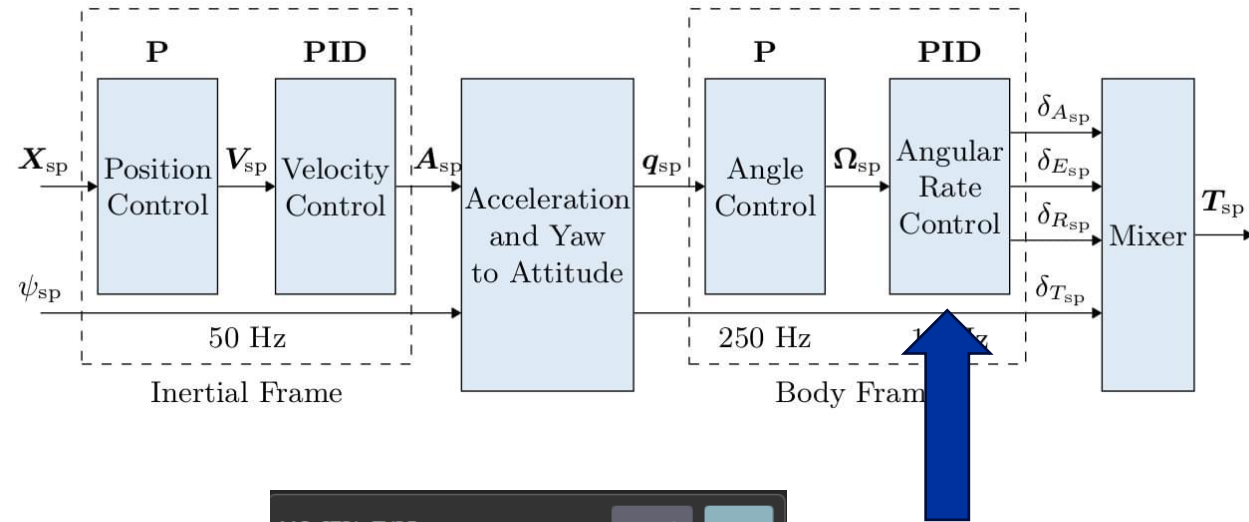


FOXGLOVE ANALYZATION TOOLS



CUSTOM RATE CONTROLLER LIBRARY IN PX4

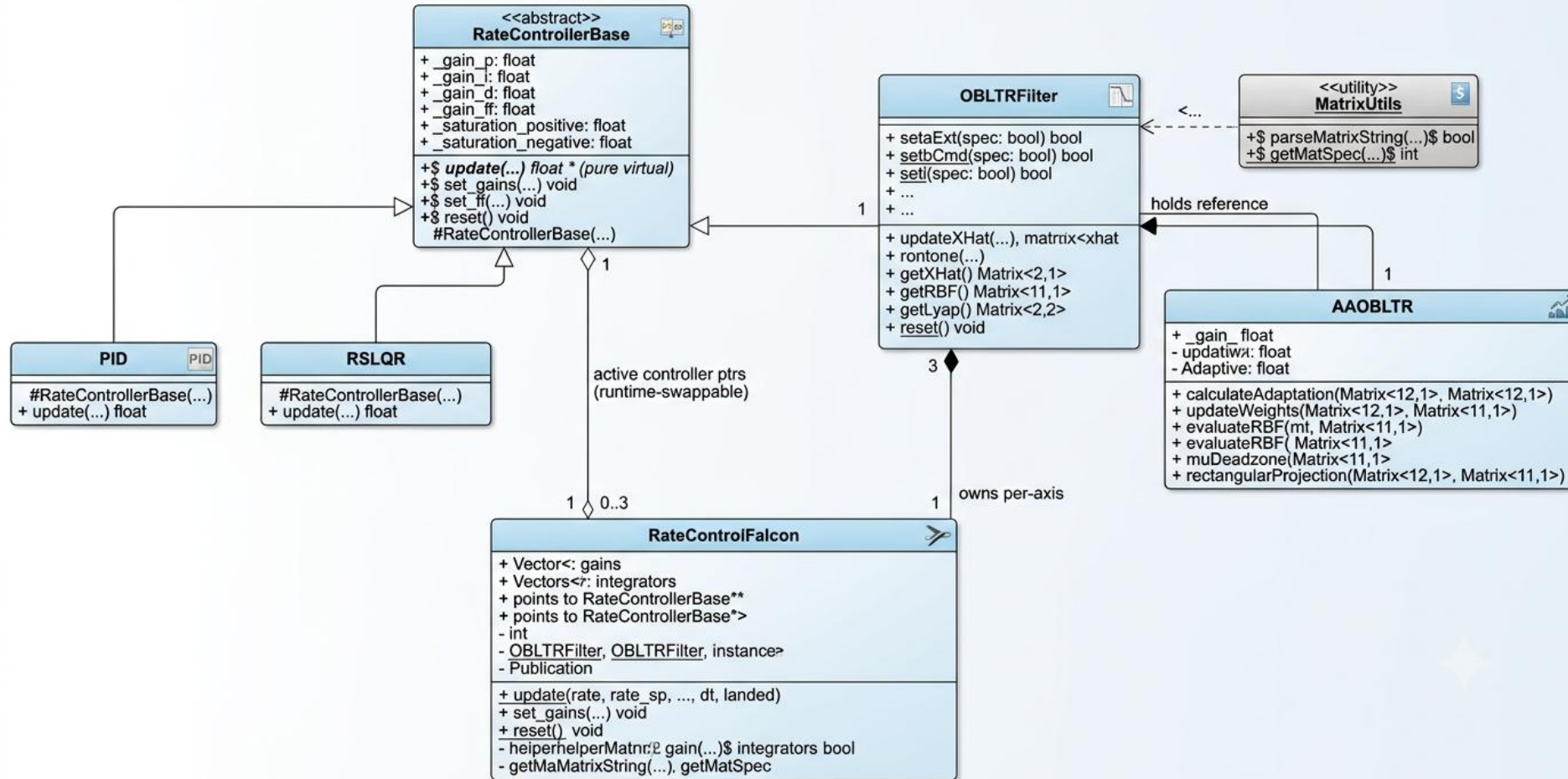
- Implemented in the angular rate control loop
- Added new uORB message types
- Extended QGC with custom tuning fields
 - Can tune in the field like with PID
 - Can not adjust Q & R



The screenshot shows the **MC_CTRL_TYPE** settings window. The **MC_CTRL_TYPE** field is set to 3. The **Reset To Default** button is visible. The description states: "Selects which rate controller algorithm to use." The options are: 0 RSLQR, 1 PX4 Vanilla PID Controller, 2 OBLTR, and 3 AAOBLTR. The default is 0. A warning message is displayed: "Warning: Modifying values while vehicle is in flight can lead to vehicle instability and possible vehicle loss. Make sure you know what you are doing and double-check your values before Save!" There is also an **Advanced settings** checkbox.

CUSTOM RATE CONTROLLER LIBRARY IN PX4

FALCON RATE CONTROL CLASS DIAGRAM



ADAPTIVE AUGMENTED OBSERVER-BASED CONTROL WITH LOOP TRANSFER RECOVERY (AAOBLTR)

- OBLTR:
 - Estimates states where the drone can't directly sense due to noise or additional sensors
 - Flies as if every state were known
- AAOBLTR:
 - Builds an OBLTR with online self-adjustment
 - Converges back to designed transient performance

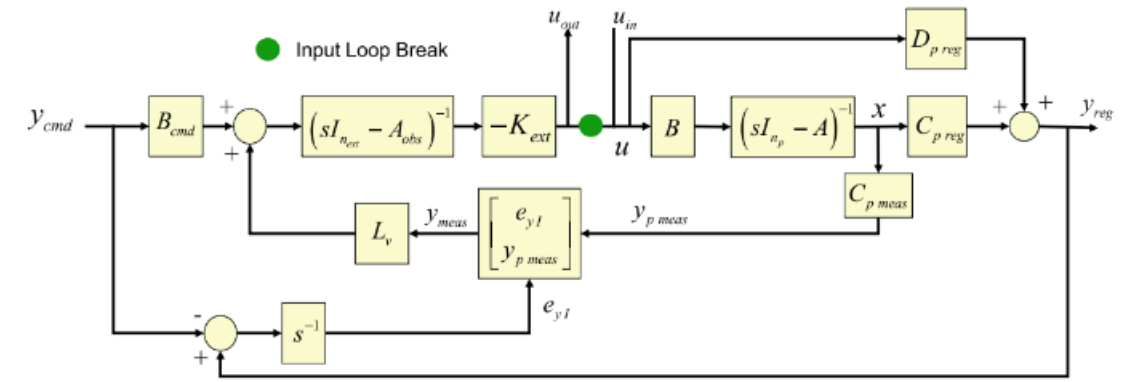
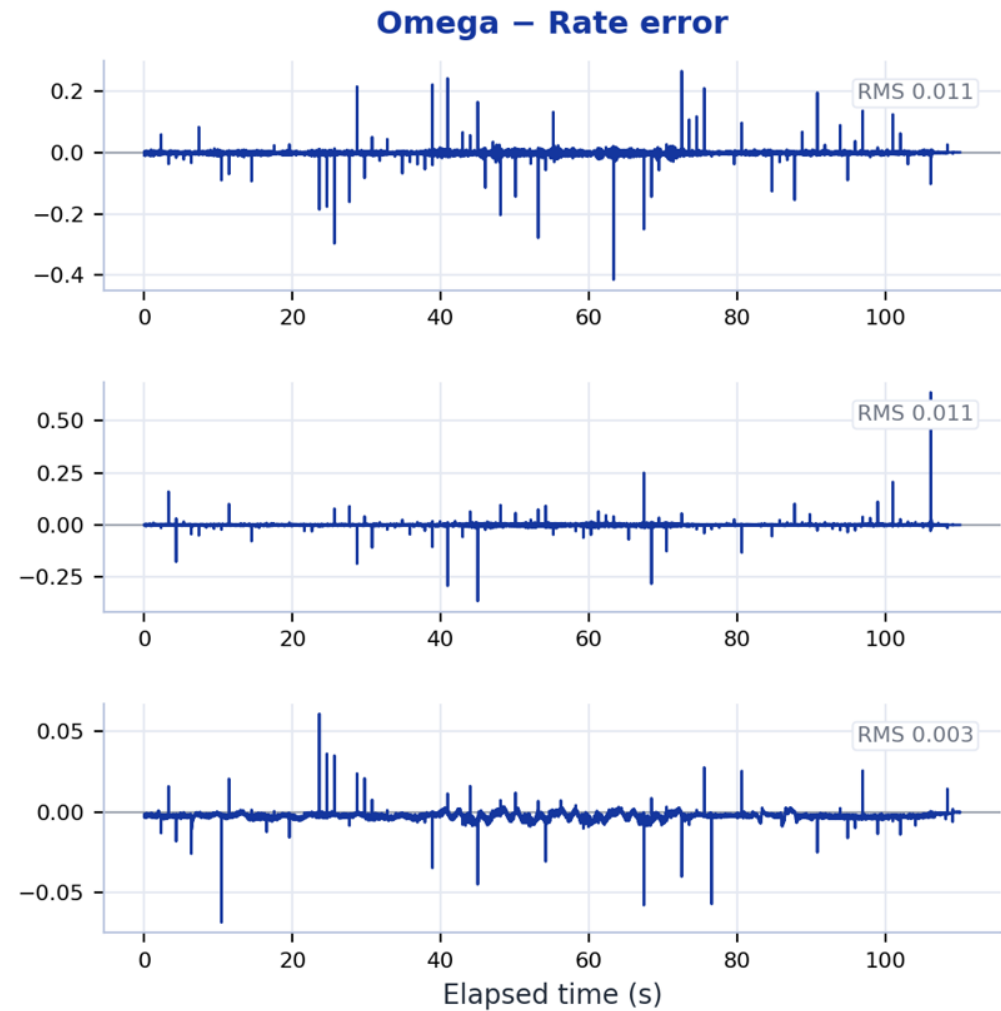
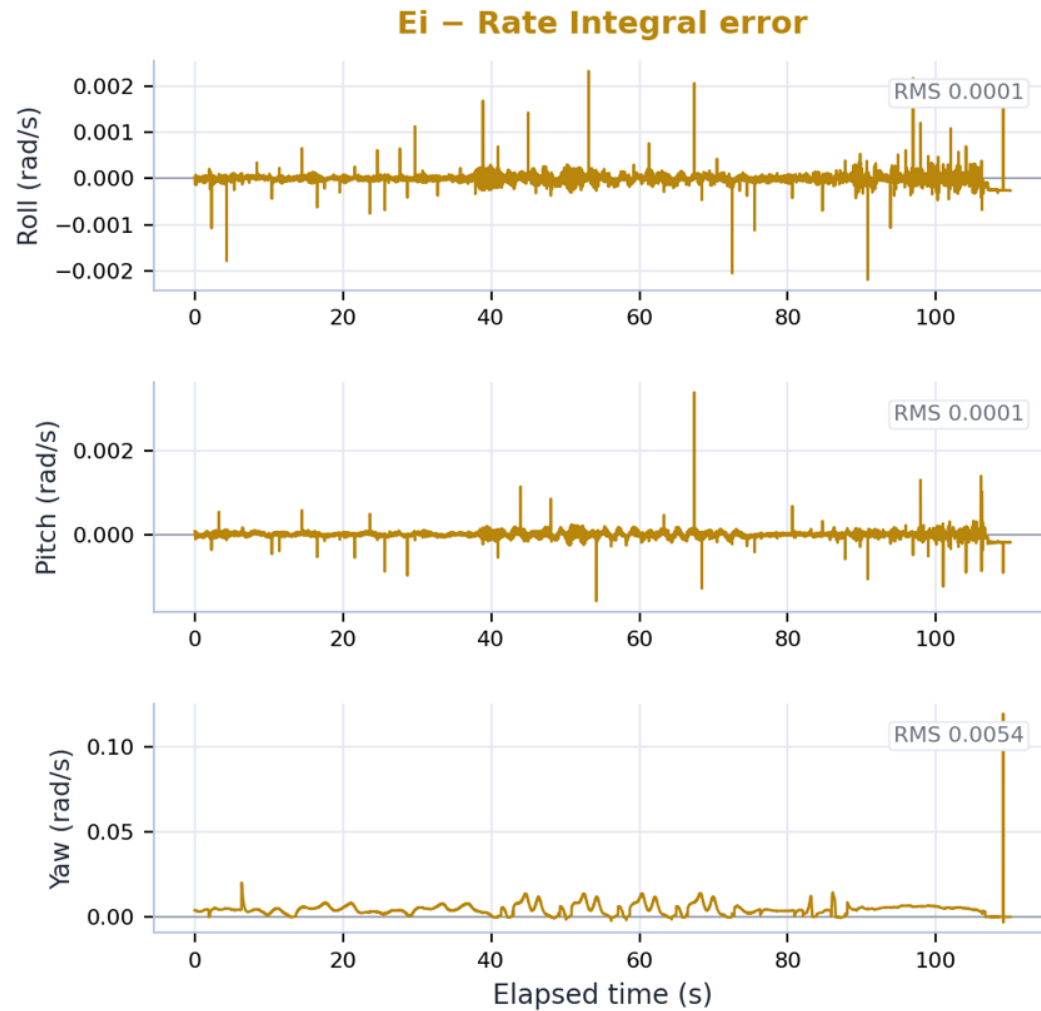


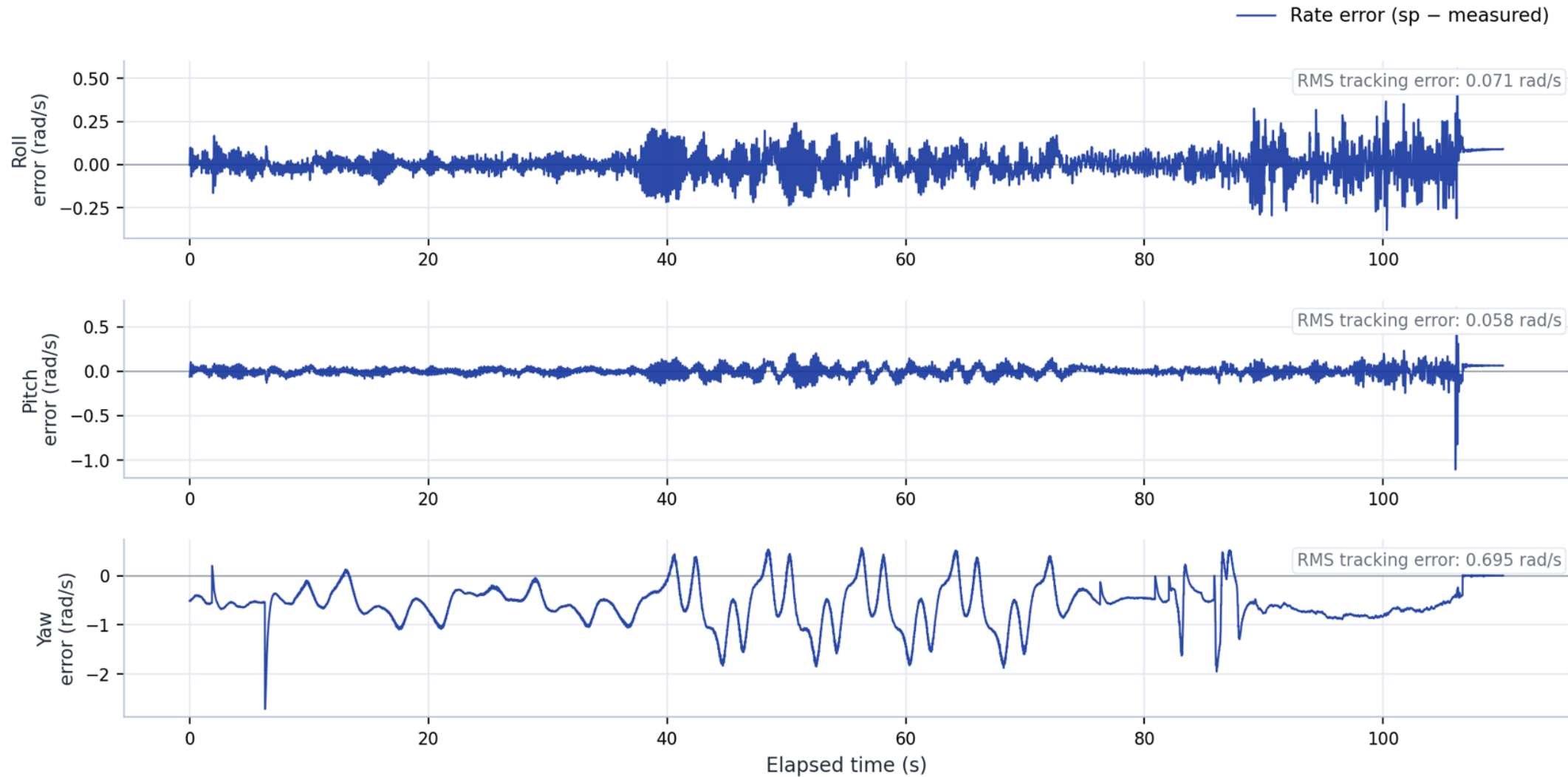
Fig. 6.13 Input loop gain calculation block diagram

TUNING OBLTR / AAOBLTR

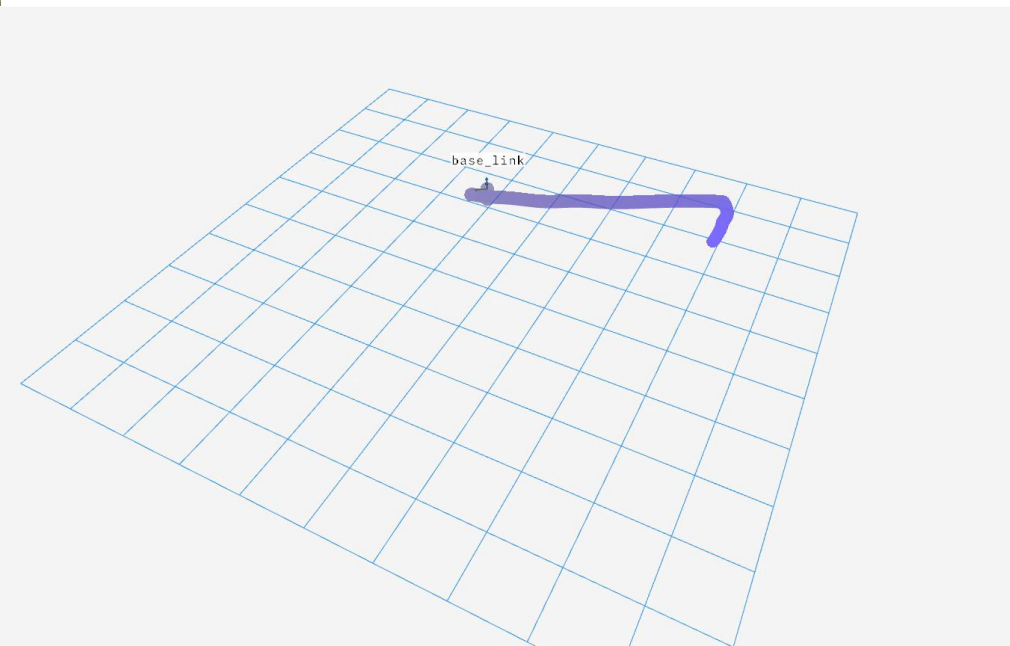
~Zero error between
estimated and real
states



OBLTR / AAOBLTR TRACKING PERFORMANCE



AAOBLTR DOING A FIGURE 8



OBSTACLE AVOIDANCE ANALYTICAL CBF SIMULATION

- Closed-form safety filter
- Validated against figure-8 mission with obstacle in path

$$\Delta \mathbf{p} = \mathbf{p} - \mathbf{o}$$

$$h = \|\Delta \mathbf{p}\|^2 - r^2$$

$$\psi_1 = 2 \Delta \mathbf{p} \cdot \mathbf{v} + \alpha_1 h$$

$$\mathbf{c} = 2 \Delta \mathbf{p}$$

$$d = 2 \|\mathbf{v}\|^2 + 2 \alpha_1 (\Delta \mathbf{p} \cdot \mathbf{v}) + \alpha_2 \psi_1$$

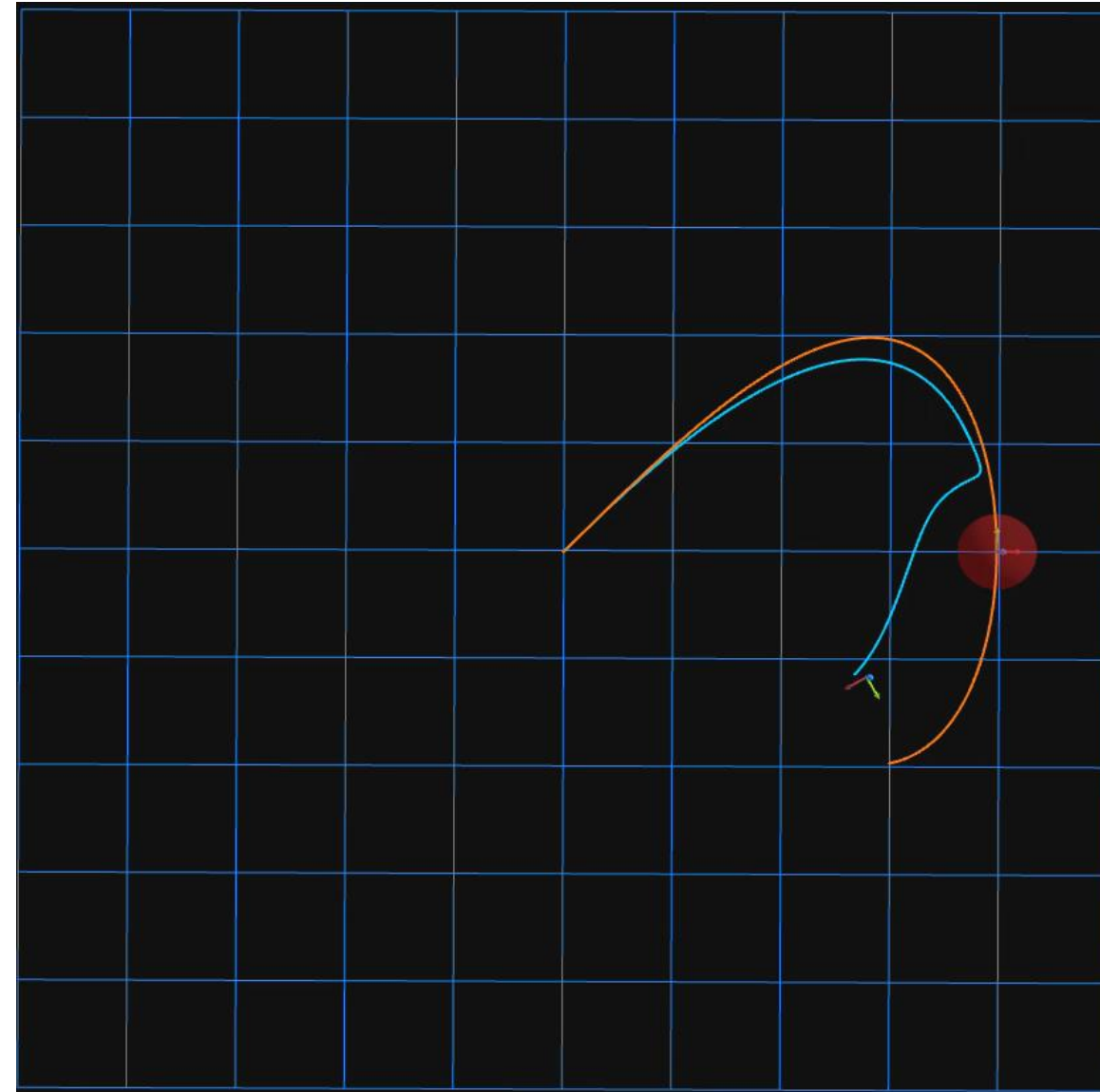
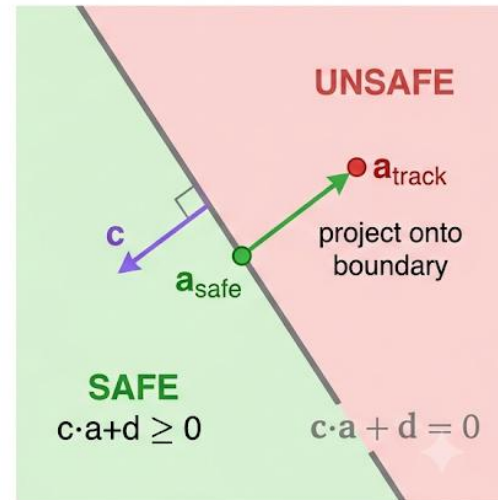
$$\text{margin} = \mathbf{c} \cdot \mathbf{a}_{\text{track}} + d$$

Solution:

$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}}, \text{margin} \geq 0$$

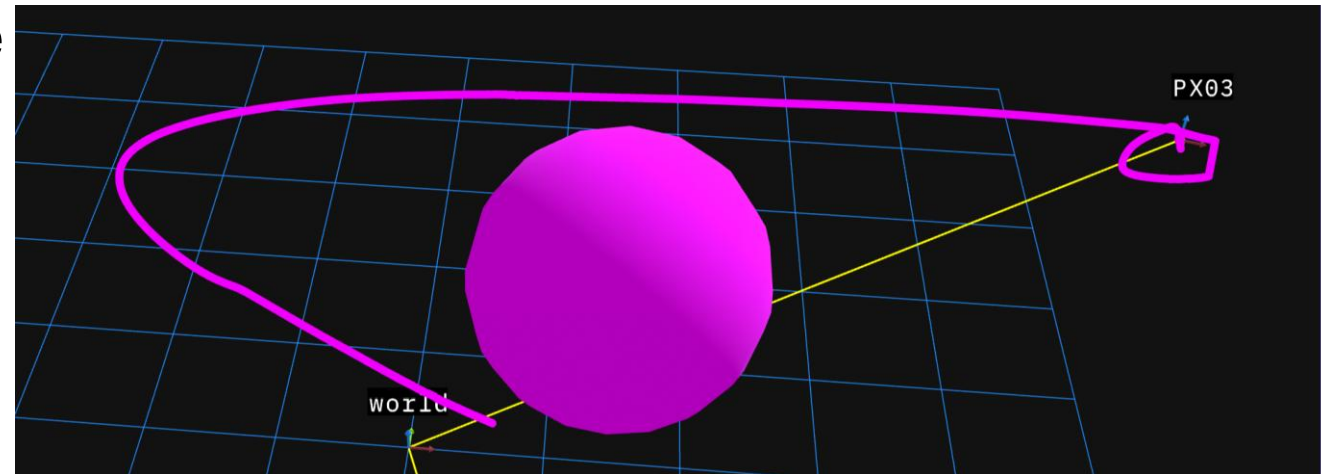
$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}} - \mathbf{c}(\text{margin}/\|\mathbf{c}\|^2), \text{margin} < 0$$

One active obstacle $\Rightarrow$ one linear inequality in $\mathbf{a}$
 $\Rightarrow$ closed form, no QP.



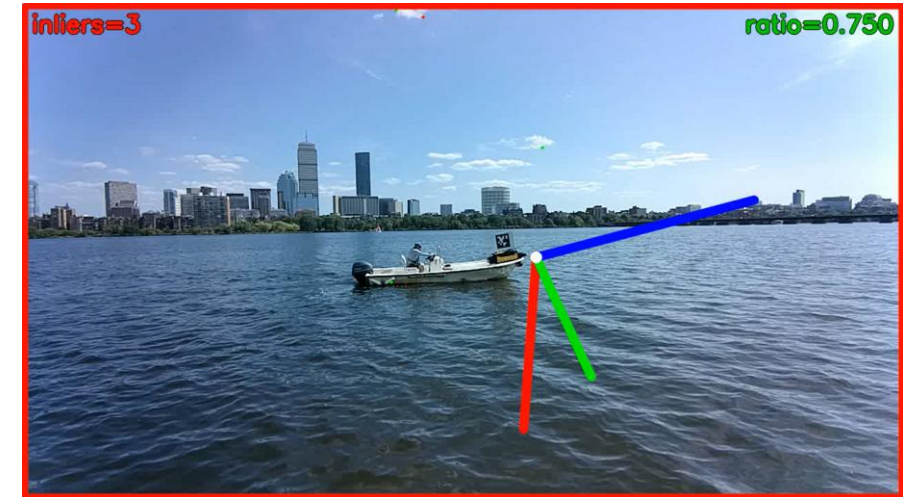
OBSTACLE AVOIDANCE ANALYTICAL CBF HARDWARE

- ROS2 namespace issues blocked goal/CBF message delivery
 - This was due to custom MIT settings
- Algorithm untested on hardware as a result
- Takeaway:
 - Need a more robust HITL pipeline



HIERARCHICAL LOCALIZATION STACK

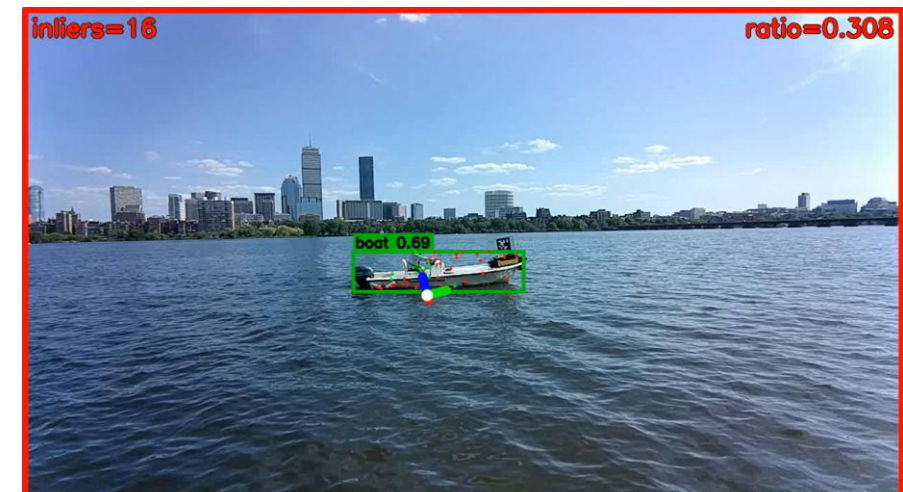
- Vision-based pose estimation for boats without relying on GPS and AprilTags
- Improved HLOC perception stack
 - Apply masking to the observed features
 - Detector: off-the-shelf YOLOv11n
- 2x the detected throughput
- ~10% Improvement in x-axis
- ~2% Improvement in heading
- ~15% Degradation in y-axis



Before



After



LESSONS LEARNED

Technical

- **Hardware testing is hard**
 - HITL is as import as SITL
 - Building reusable launch tooling to cut setup time
- Steep ramp-up on advanced controllers
- Analyzation tools are essential

Professional

- First time with hard deadlines
- Fast-paced, experienced team

People & Culture

- In-office connections matter
- Great work culture boosts productivity



Thank You To:



Intern Coordinators:

- Katie Martins
- Madelyn Focaracci
- Aidan Green
- Preston Edwards

The FALCON Team:

- Max Greene
- Jeffrey Davis
- Marcel Menner
- Spencer Folk
- Frank Gonzalez
- Trevor Houchens
- Leo Ott

Our Managers:

- Sildomar Monteiro
- Douglas Lipinski

Our Mentors:

- Christopher Walter
- Amanda Vollum
- All of AMA

Summary of Contributions



- Extended FALCON controllers and associated missions to quadrotors:
 - RSLQR, OBLTR, AAOBLTR on PX4 firmware
 - Relative station keeping with velocity control
- Designed & implemented analytical CBFs for theoretically guaranteed safe guidance
 - Obstacle avoidance
 - Relative station keeping
- Live DARPA Demonstration: use of FALCON controllers on an aerial platform

CONNECT WITH US ON LINKEDIN



Amikosh Dube



Suchitha Channapatna